

Trainings ▾

Preparatory Steps

Automotive with CM870

- Remove the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).
- Remove the cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/10/10-008-002-tr.html>).
- Remove the automatic belt tensioner (water pump). Refer to Procedure 008-080 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).



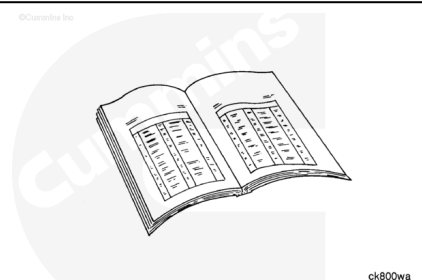
Automotive with CM570

- Remove the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).
- Remove the water pump belt. Refer to Procedure 008-003 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).
- Remove the alternator belt Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).



Industrial CM570 and Power Generation without CM570

- Remove the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).
- Remove the fan and water pump belt. Refer to Procedure 008-003 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).
- Remove the alternator drive belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Remove the fan hub and the fan hub support. Refer to Procedure 008-088 (</qs3/pubsys2/xml/en/procedures/10/10-008-088-tr.html>).

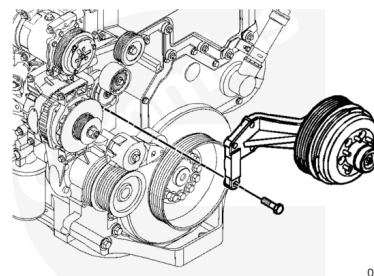


Remove

Automotive with CM870

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

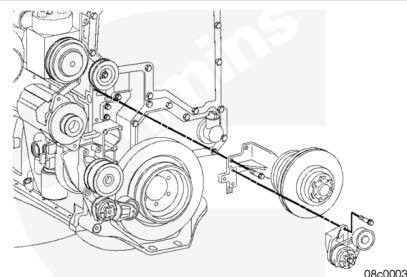


Remove the mounting capscrews and the fan hub assembly.

Automotive with CM570

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



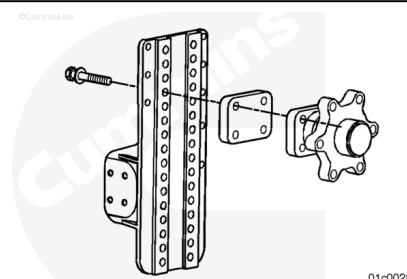
Remove alternator belt tensioner and bracket.

Remove the capscrews and the fan hub assembly.

Industrial CM570 and Power Generation without CM570

Remove the fan hub mounting capscrews.

Remove the fan hub from the fan support.



Clean and Inspect for Reuse

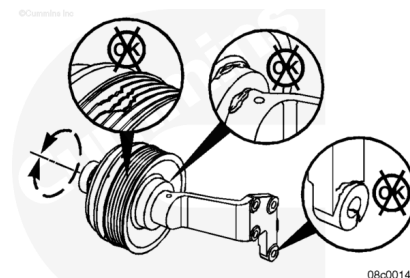
Automotive with CM870

Inspect the fan hub for the following:

- Freedom of rotation
- Cracks
- Grease seal leakage.

Repair or replace the fan hub if the fan hub does **not** rotate freely or if there is evidence of cracks or grease seal leakage.

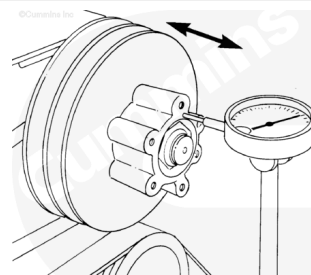
Refer to the OEM service literature for repair of the fan hub.



Measure the fan hub end clearance play. The end clearance is dependent upon the fan hub style.

Replace the fan hub if the end clearance is **not** within the manufacturer's specifications.

Refer to OEM service manual for additional inspection steps.

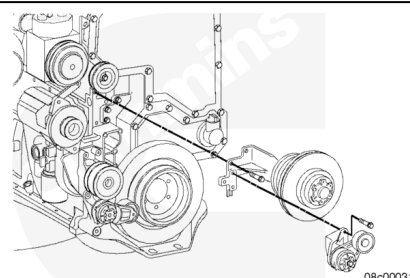


Automotive, Industrial with CM570 and Power Generation without CM570

Inspect the fan hub for the following:

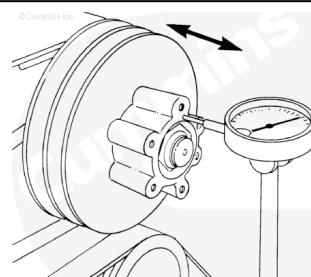
- Freedom of rotation
- Cracks
- Grease seal leakage.

Repair or replace the fan hub if the fan hub does **not** rotate freely or if there is evidence of cracks or grease seal leakage.



Measure the fan hub end clearance. The end clearance is dependent upon the fan hub style.

Replace or repair the fan hub if the end clearance is **not** within specifications.



Fan Hub End Clearance		
mm		in
0.08	MIN	0.003
0.25	MAX	0.009

Disassemble

Automotive, Industrial with CM570 and Power Generation without CM570

Note : Disassemble fan hub only if fan hub end clearance is larger than exceeds specification.

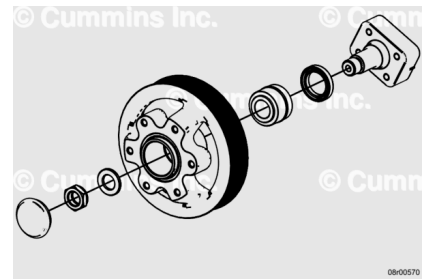
Remove the fan hub cover.

Remove the lock nut and plain washer.

Remove fan hub from fan spindle.

Remove and discard the roller bearing from fan hub.

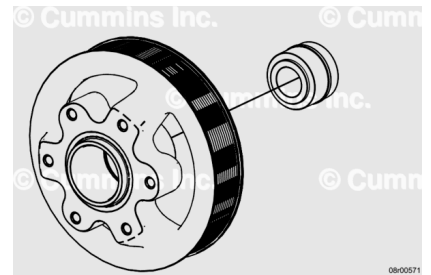
Remove and discard the oil seal.



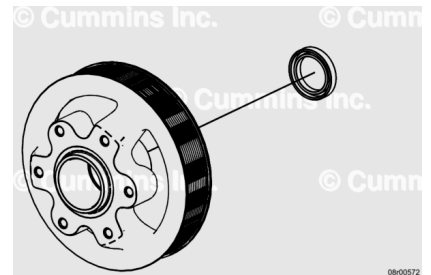
Assemble

Automotive, Industrial with CM570 and Power Generation without CM570

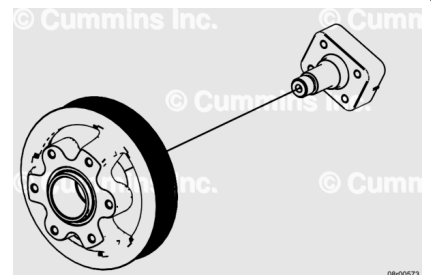
Press a new roller bearing into the fan hub so that the captured retaining ring sits towards the rear of the fan hub. Verify the retaining ring seats into the groove of the fan hub.



Press a new oil seal into the fan hub.



Install the fan hub assembly to the fan spindle.

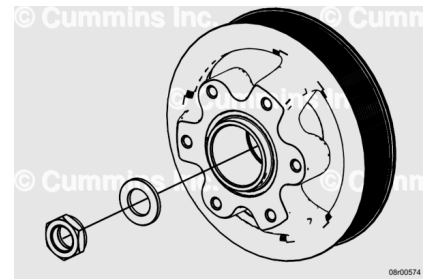


Install the plain washer and lock nut.

Tighten the lock nut.

Torque Value: 203 n•m [150 ft-lb]

Install the fan hub cover.

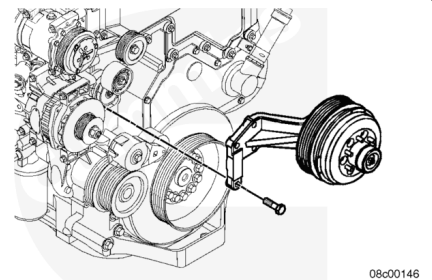


Install

Automotive with CM870

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



Install the fan hub assembly and the capscrews.

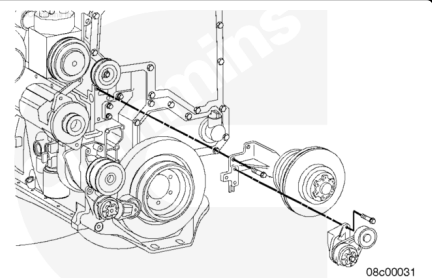
Tighten the capscrews.

Torque Value: 113 n•m [83 ft-lb]

Automotive with CM570

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this assembly.



Install the fan hub and the capscrews.

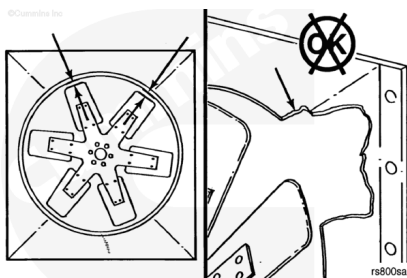
Tighten the capscrews.

Torque Value: 90 n•m [66 ft-lb]

Install the belt tensioner bracket.

Industrial CM570 and Power Generation without CM570

The fan hub **must** be installed in the correct location on the fan support to allow proper clearance between the fan and fan shroud. Refer to the OEM service manual.

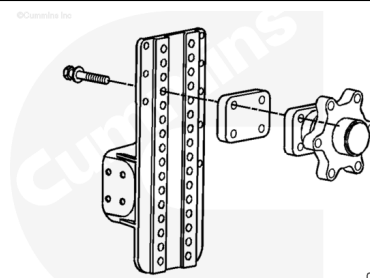


Make certain the correct fan center is used for the application.

Install the fan hub on the fan support using four capscrews and washers.

Tighten the capscrews.

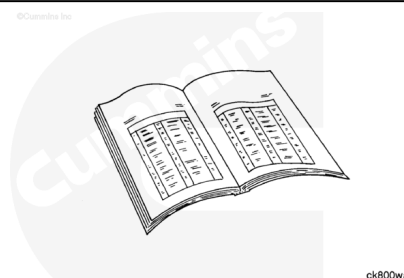
Torque Value: 90 n·m [66 ft·lb]



Finishing Steps

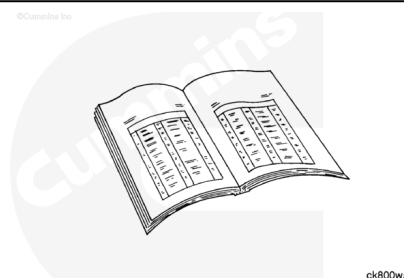
Automotive with CM870

- Install the automatic belt tensioner (water pump). Refer to Procedure 008-080 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Install the cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/10/10-008-002-tr.html>).
- Install the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).



Automotive with CM570

- Install the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/10/10-013-005.html>).
- Install the water pump belt. Refer to Procedure 008-003 (</qs3/pubsys2/xml/en/procedures/10/10-008-003.html>).
- Install the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/10/10-008-040.html>).



Industrial CM570 and Power Generation without CM570

- Install the fan hub and fan hub support. Refer to Procedure 008-088 (/qs3/pubsys2/xml/en/procedures/10/10-008-088.html).
- Install the alternator drive belt. Refer to Procedure 013-005 (/qs3/pubsys2/xml/en/procedures/10/10-013-005.html).
- Install the fan and water pump belt. Refer to Procedure 008-003 (/qs3/pubsys2/xml/en/procedures/10/10-008-003.html).
- Install the cooling fan. Refer to Procedure 008-040 (/qs3/pubsys2/xml/en/procedures/10/10-008-040.html).



Last Modified: 10-Aug-2022
